

Models and Tools for Cyber-Physical Systems

Chapter 7: Dynamical Systems - Part 2

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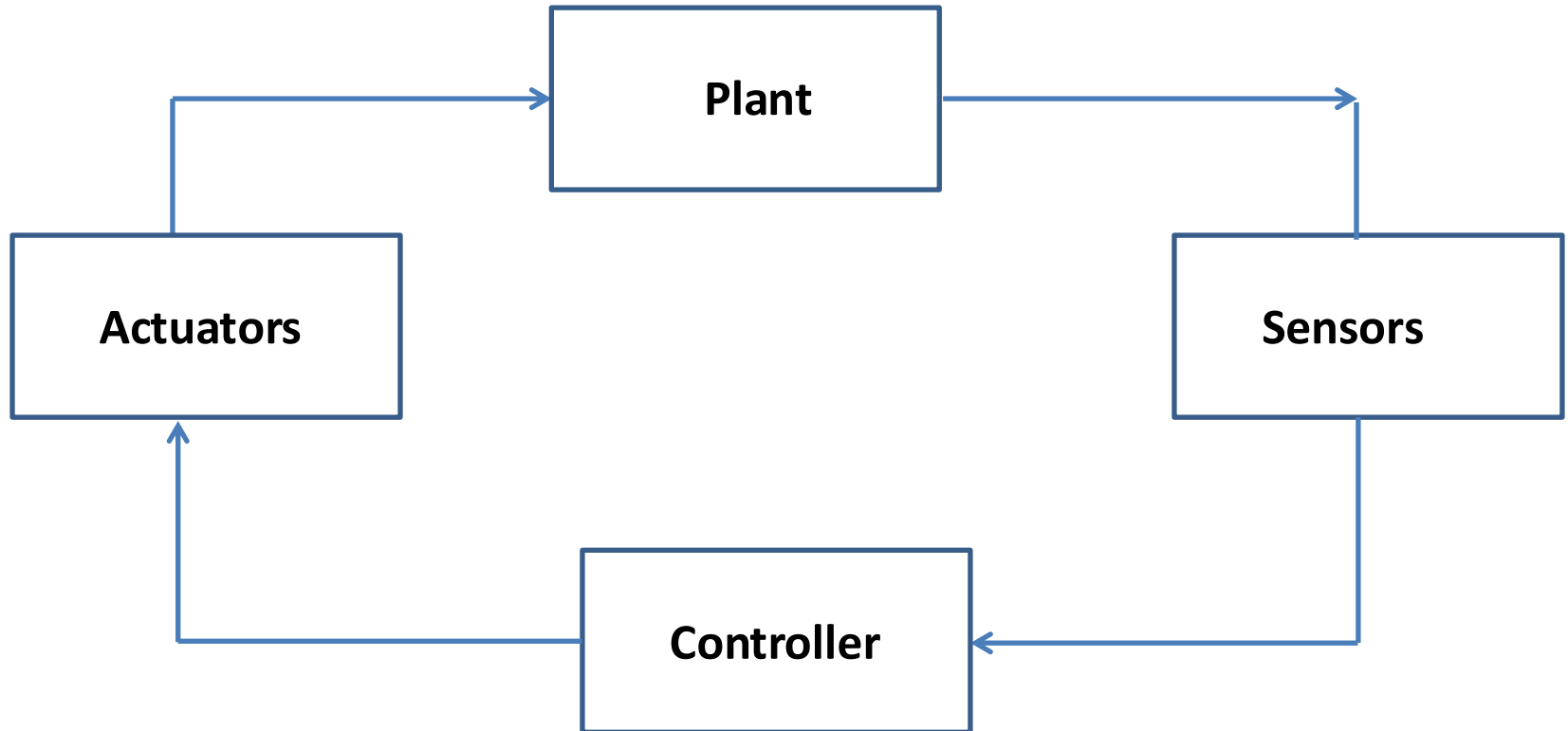
Slides courtesy of Rajeev Alur
alur@cis.upenn.edu

Instability

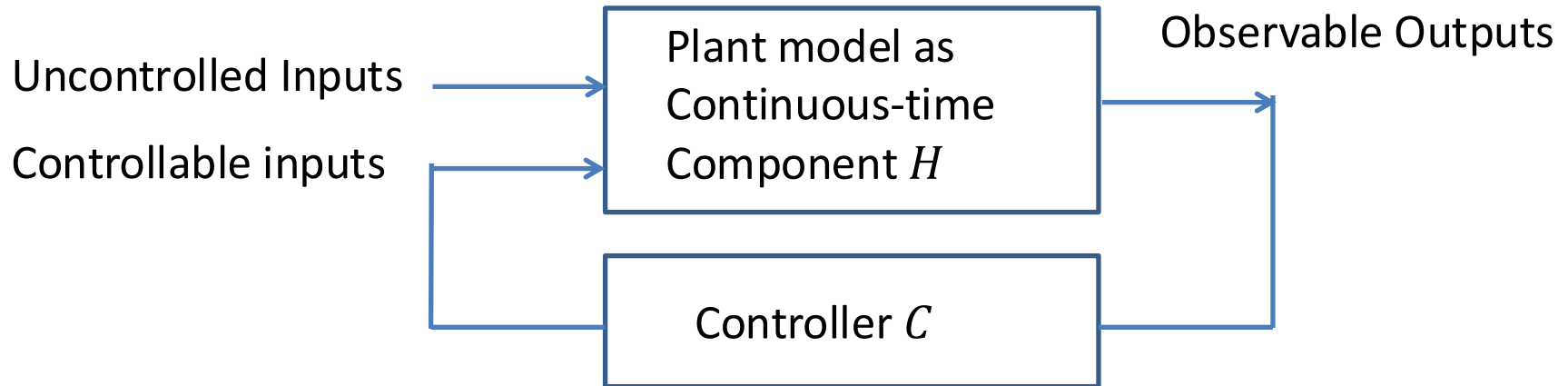
- ❑ If a continuous system is stable, it is considered to be correct.
- ❑ But what if a continuous system is unstable? Should we say to the mechanical engineers that they did a bad job at designing the dynamical system?
- ❑ Sometimes you can change the physical design to make it stable (towing a trailer example: https://youtu.be/6mW_gzdh6to?t=6), but this is not always possible.
- ❑ A Segway is an inverted pendulum (which is the unstable equilibrium point!)
- ❑ Solution: control software! (next lecture)



Traditional Feedback Control Loop

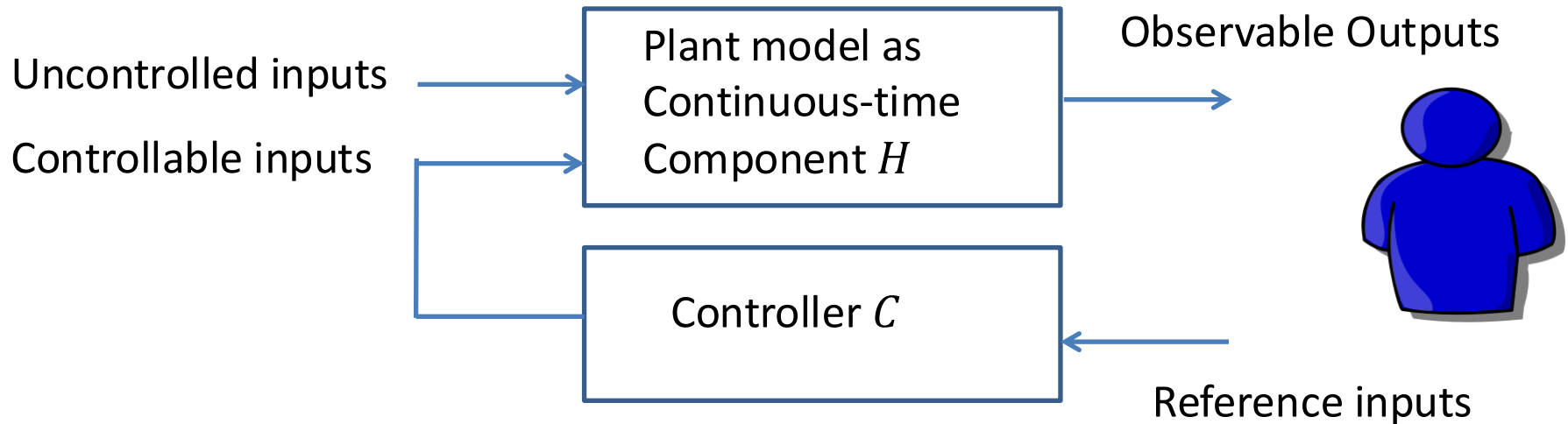


Control Design Problem



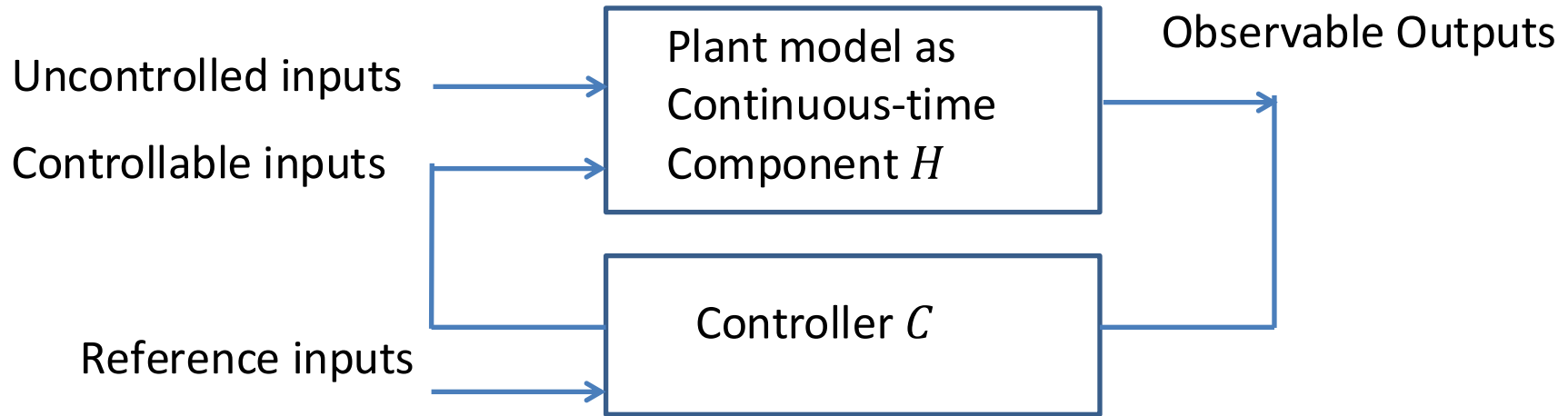
- ☐ Design a controller C so that the composed system $C||H$ is stable
- ☐ Is there a mathematical way to check when a system is stable?
- ☐ Is there a way to design C so that $C||H$ is stable ?
- ☐ Yes, if the plant model is **linear**! → Material for advanced Master's course.
- ☐ In this course, we will stick to analyzing controllers by simulation.

Open Loop Controller



- ❑ Plant outputs not fed to the controller
 - Benefit: Sensors not needed (less expensive)
- ❑ Controller simply maps reference inputs to controllable inputs
 - Knowledge of plant dynamics hard-coded in this algorithm
- ❑ Human intervention necessary to maintain acceptable performance

Closed Loop Feedback Controller



- ❑ Controller adjusts controllable inputs in response to observed outputs
 - Can respond better to variations in disturbances
 - Performance depends on how well outputs can be measured
- ❑ Two control design techniques
 - Mathematical based on theory of linear systems
 - PID controllers (widely used in practice)

Aircraft Model (Simplified)

Design problem: What torque should the tail apply to keep the aircraft from spinning?

Yaw = ψ

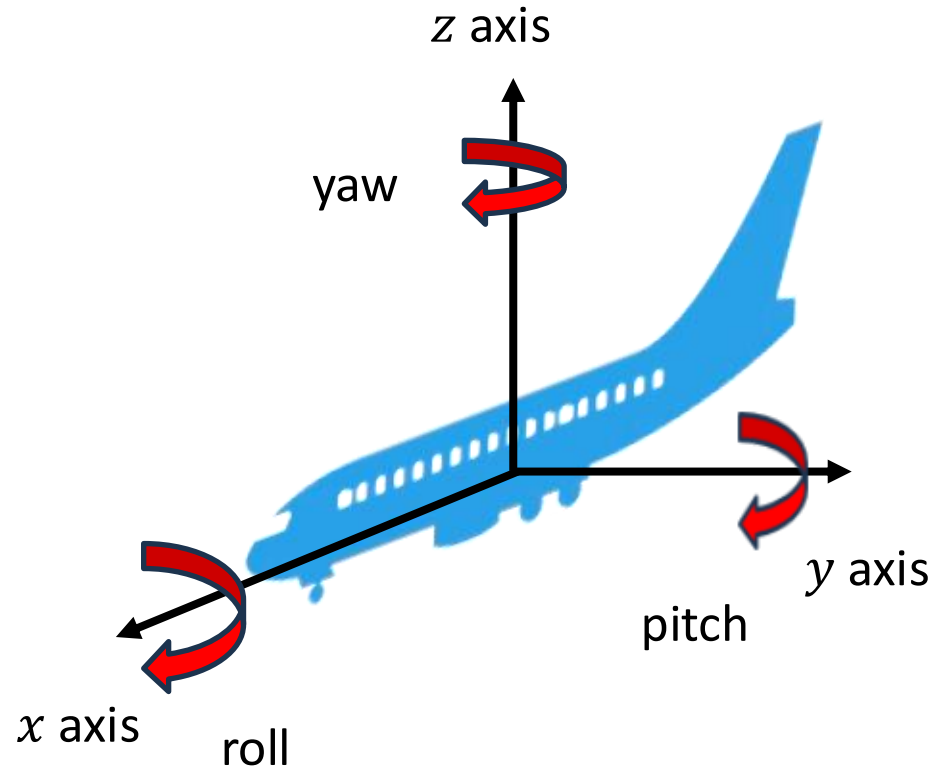
Spin (rate of change of yaw) = s

Torque by rotor: T

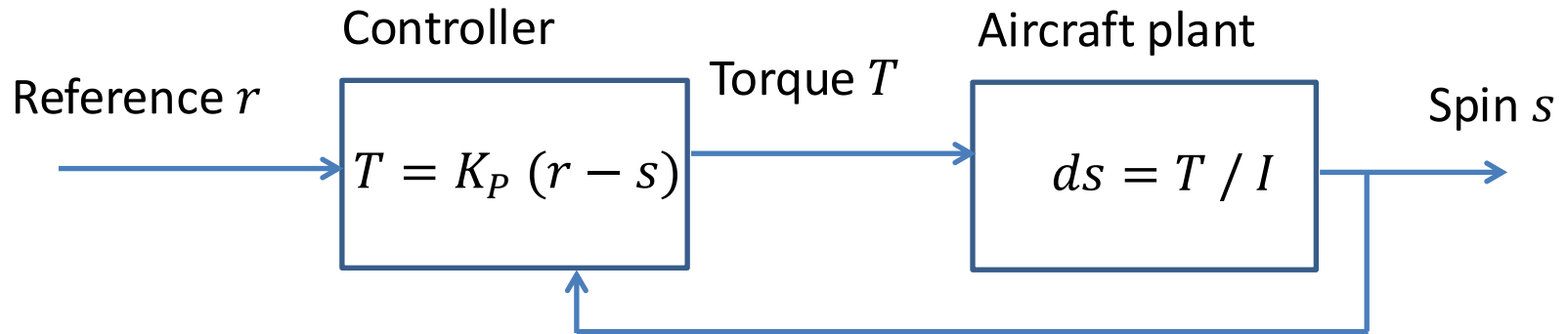
Moment of inertia: I

Equation of motion:

$$ds/dt = T / I$$

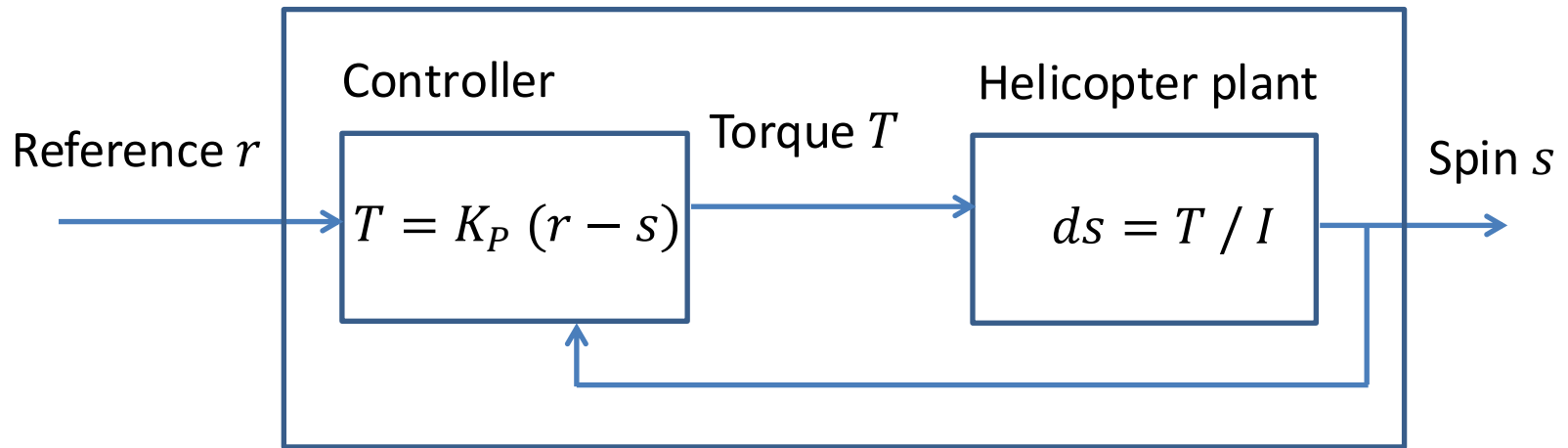


Feedback Controller for Helicopter Model



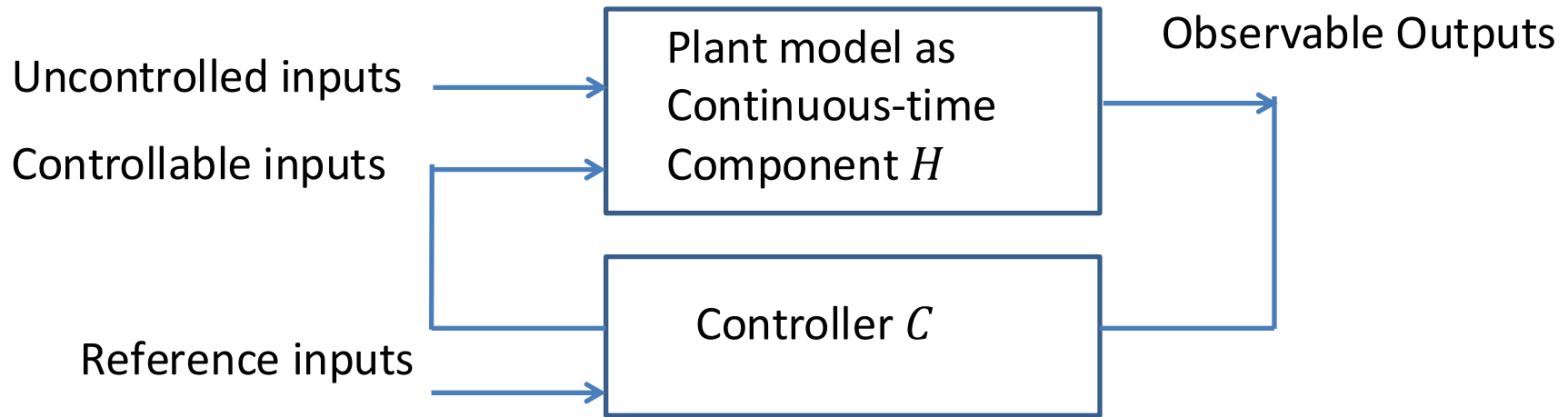
- ❑ Design controller so that composed system is stable
- ❑ Error $e = (r - s)$: difference in desired value and observed output
- ❑ Proportional controller: Its output is proportional to this error
- ❑ Constant K_P : Proportional Gain
- ❑ Note that the direction of torque changes with sign of the error

Stabilizing Controller for Helicopter Model



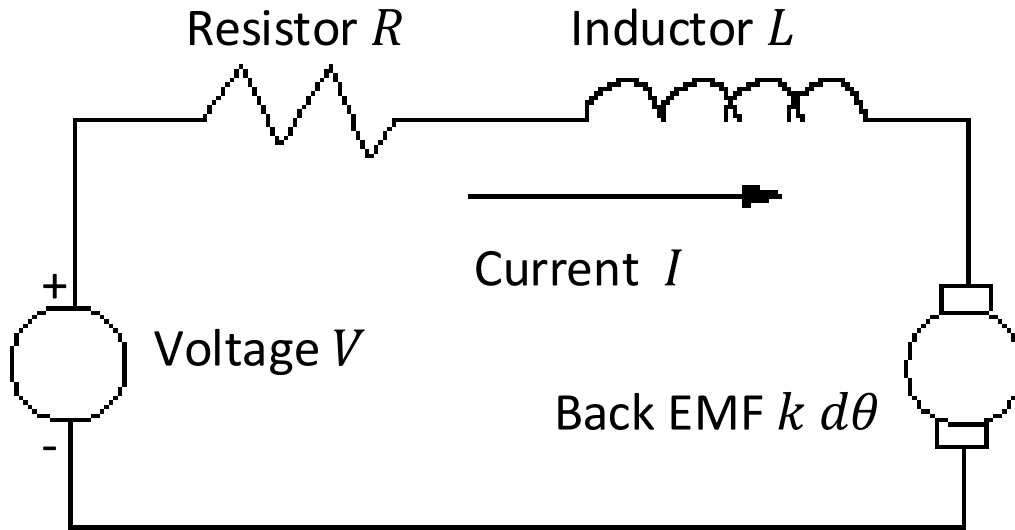
- ❑ Dynamics of the composed system:
$$ds/dt = K_P (r - s) / I$$
- ❑ When is this system asymptotically stable ?
- ❑ If the coefficient $-K_P/I$ is negative
- ❑ Control design: choose a positive gain constant K_P
 - Rate of convergence depends on its magnitude

PID Controllers



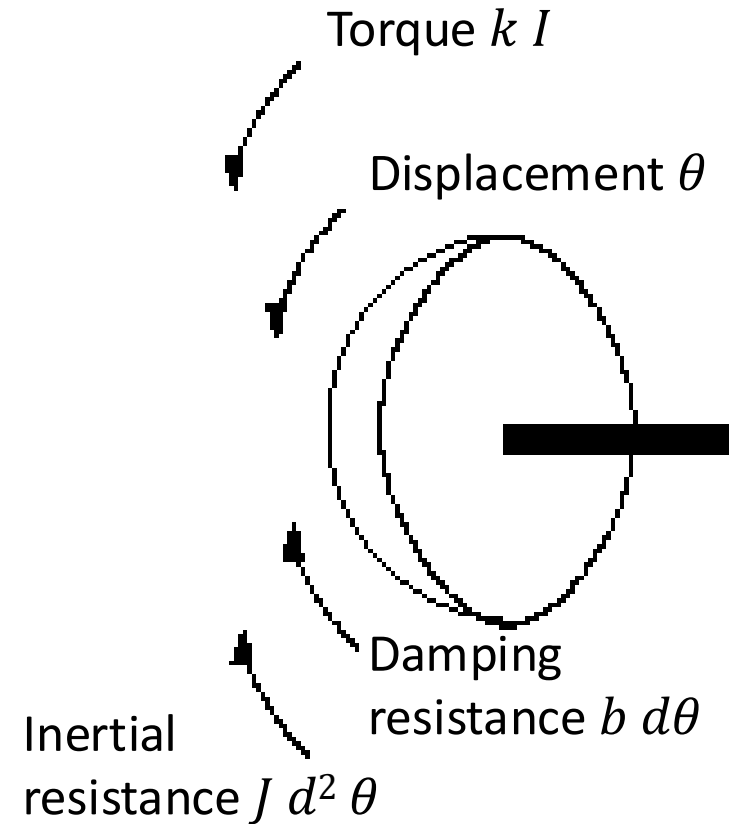
- ❑ Strategy for designing controllers that is widely used in practice
- ❑ Error = Reference Inputs - Observable Outputs
- ❑ Control output is sum of 3 terms:
 - Term proportional to error
 - Integral term to handle cumulative error
 - Derivative term in response to rate of change of error

DC Motor



Laws of electrical circuits:

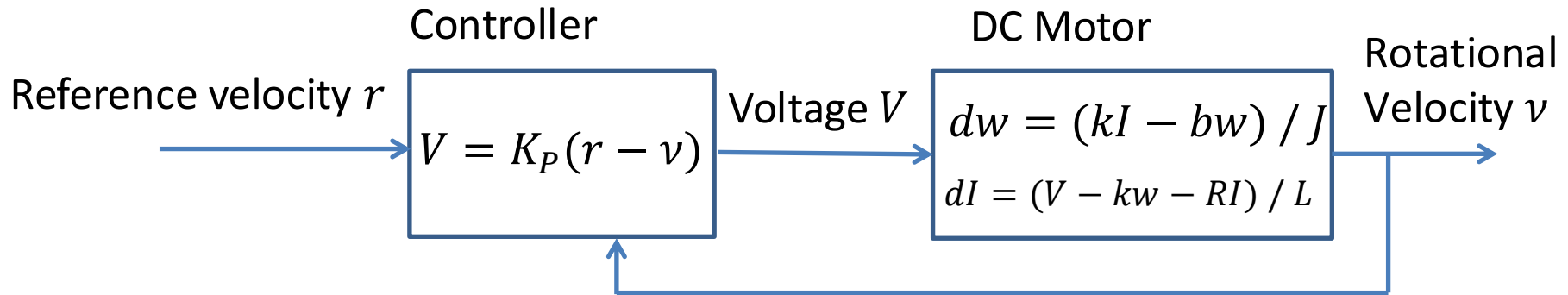
$$L \frac{dI}{dt} + R I + k d\theta/dt = V$$



Laws of motion for the shaft:

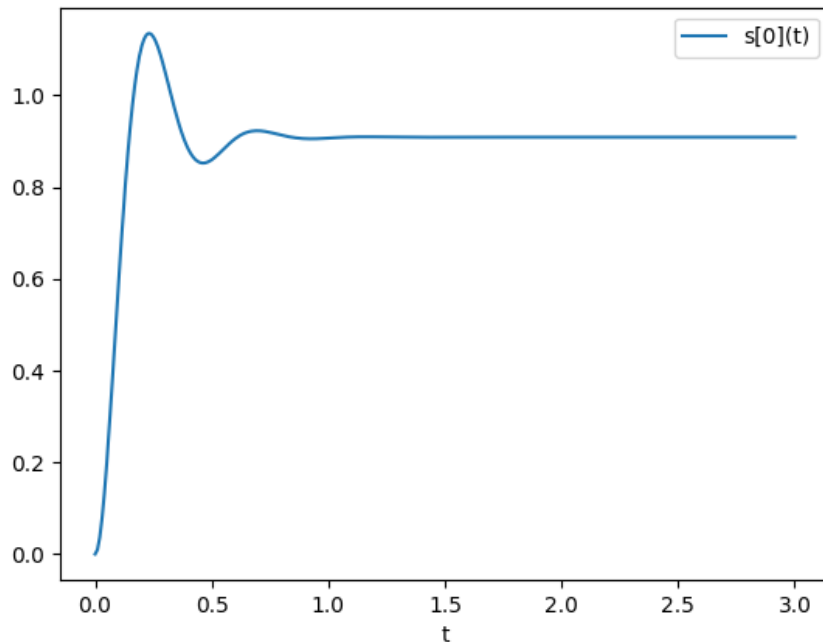
$$J d^2 \theta/dt^2 + b d\theta/dt = k I$$

Proportional Controller for DC Motor



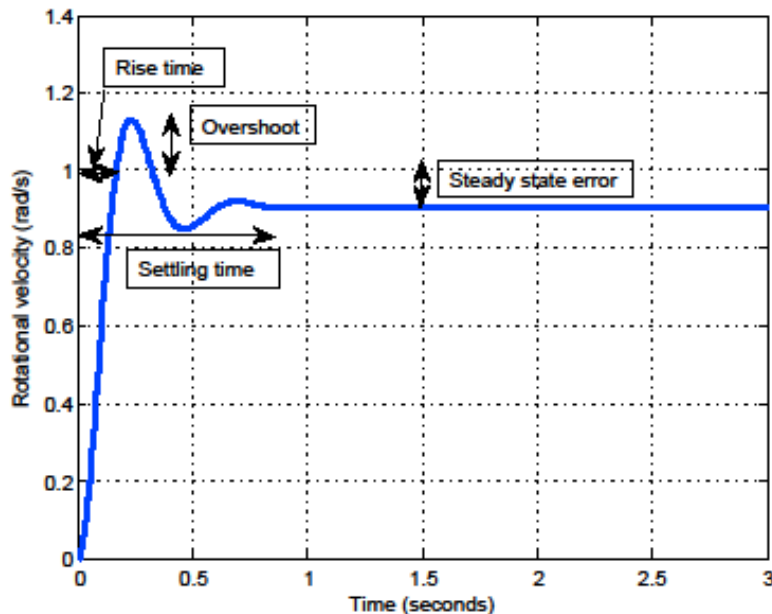
- ❑ DC Motor modeled as a linear system with 2 state variables, 1 input variable, and 1 output variable
- ❑ Feedback controller observes rotational velocity, and adjusts voltage to make it equal to desired rotational speed r
- ❑ First attempt: Proportional controller (P controller)

Step Response of P Controller



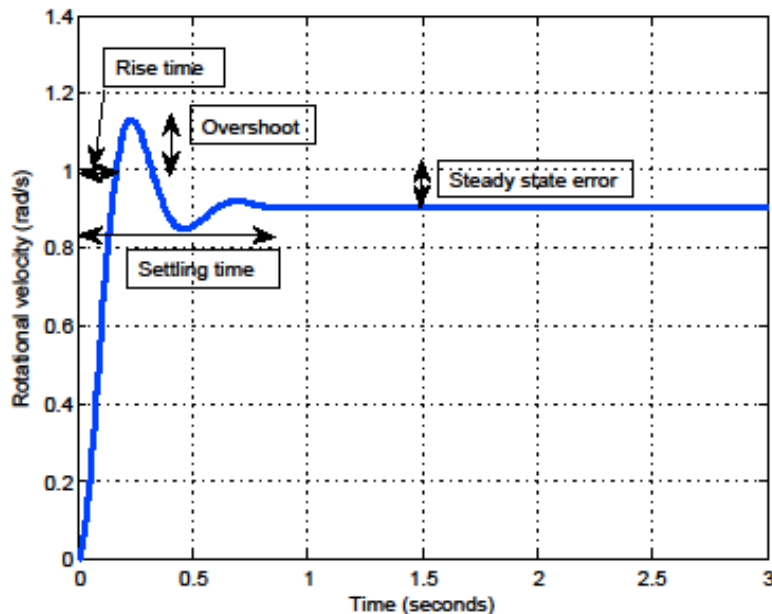
- ❑ Step response: How will system output change if at time 0, with $v = 0$, we change reference input r to 1?
- ❑ Plotted using Python (see book for values of various parameters)
- ❑ Beyond stability and convergence, what are desired characteristics of the response?

Characteristics of the Step Response



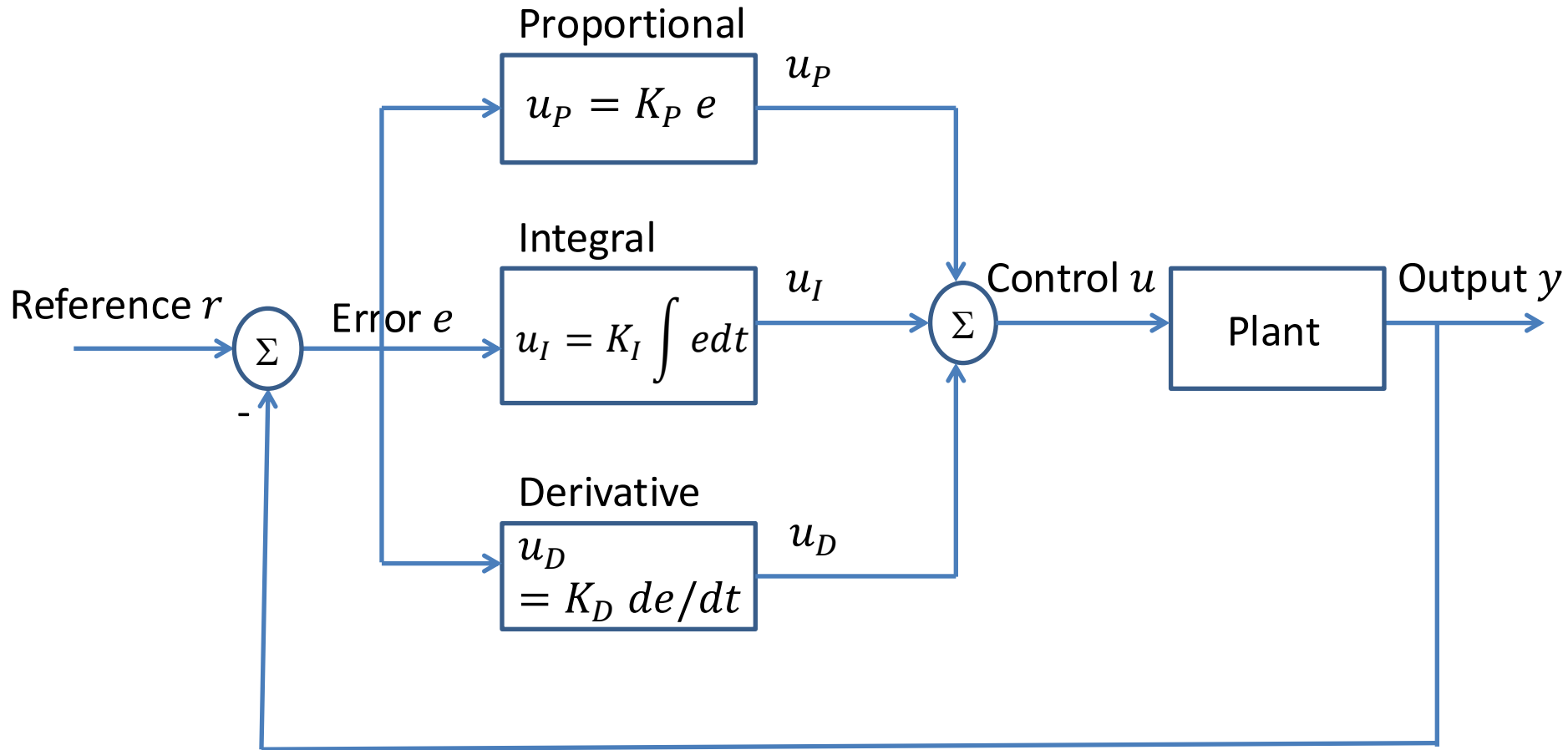
1. *Overshoot*: Difference between maximum output value and reference value (11% in this plot)
2. *Rise Time*: Time at which the output value crosses reference value (0.15sec in this plot)
3. *Settling Time*: Time at which output value reaches steady-state value (0.8sec in this plot)
4. *Steady State Error*: Difference between steady-state output value and reference (10% in this plot)

Improving the Step Response



- ❑ Performance of the P-controller depends on the value of the proportional gain constant K_P
- ❑ What happens if we increase it?
- ❑ Rise time decreases, but overshoot increases
- ❑ Steady-state error remains!
- ❑ Solution: Use Integral and Derivative Gains

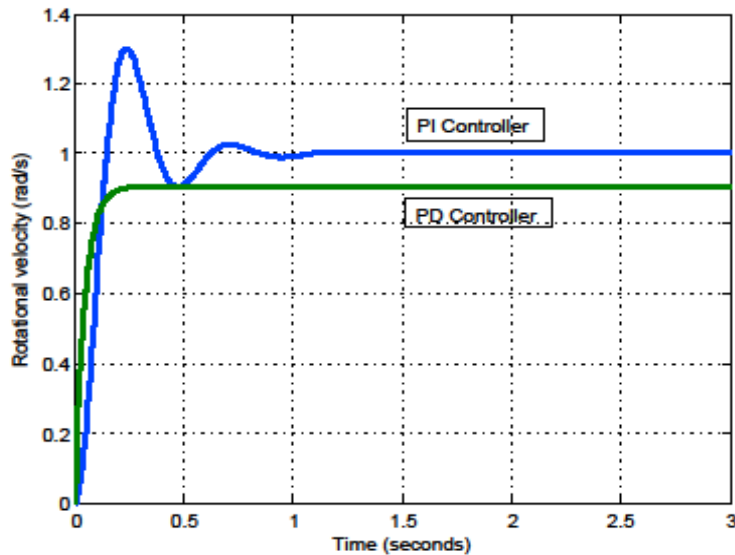
PID Controller



PID Controller

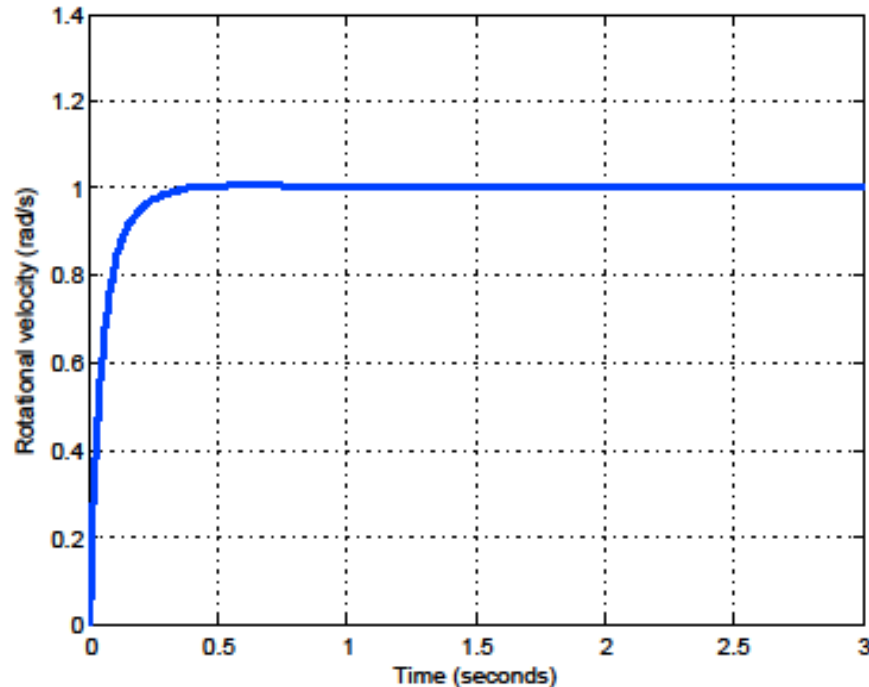
- ❑ If $e(t)$ is the error signal, then the output $u(t)$ of the PID controller is sum of 3 terms:
 - Proportional term: $K_P e(t)$, K_P is called proportional gain (response to current error)
 - Integral term: $K_I \int e dt$, K_I is integral gain (response to error accumulated so far)
 - Derivative term: $K_D d/dt e(t)$, K_D is derivative gain (response to current rate of change of error)
- ❑ Special cases of controllers: P, PD, PI

PI and PD Controllers for DC Motor



- ❑ PI Controller: adding integral term to proportional controller gets rid of steady state error
 - Overshoot, rise time, settling time increase
- ❑ PD controller: adding derivative term to proportional controller gets rid of overshoot
 - Steady state error remains

PID Controller for DC Motor



Excellent performance on all metrics: $K_P = 100$, $K_D = 10$, $K_I = 200$
Small rise time, settling time, negligible steady state error, no overshoot

Designing PID Controllers

- ❑ What are the effects of changing the gain constants K_P , K_D , K_I ?
- ❑ Broad co-relationships well understood
- ❑ Control toolboxes allow automatic tuning of parameters
- ❑ PID controllers seem to work well even when the actual system differs significantly from the plant model
 - Computation of control output depends only on the measured error, and not on the model!
- ❑ Good video reference:
 - ❑ <https://www.youtube.com/watch?v=wkfEZmsQqiA&t=34s>

Cruise Control Exercise

From exercise sheet 9, do

- ❑ Exercise 1 - cruise control for a car model.